

Perceptron and Logistic Regression

Perceptron Learning Algorithm

The Perceptron Learning Algorithm (PLA) is a supervised learning algorithm widely employed for binary classification tasks. Its primary objective is to determine a decision boundary that effectively separates the two classes in the dataset. This algorithm belongs to the class of discriminative classification methods as it focuses on modeling the boundary between classes instead of characterizing the underlying probability distribution of each class.

$$D = \{(x_1, y_1), \dots, (x_n, y_n)\}$$
$$x_i \in \mathbb{R}^d \text{ and } y_i \in \{+1, -1\}$$

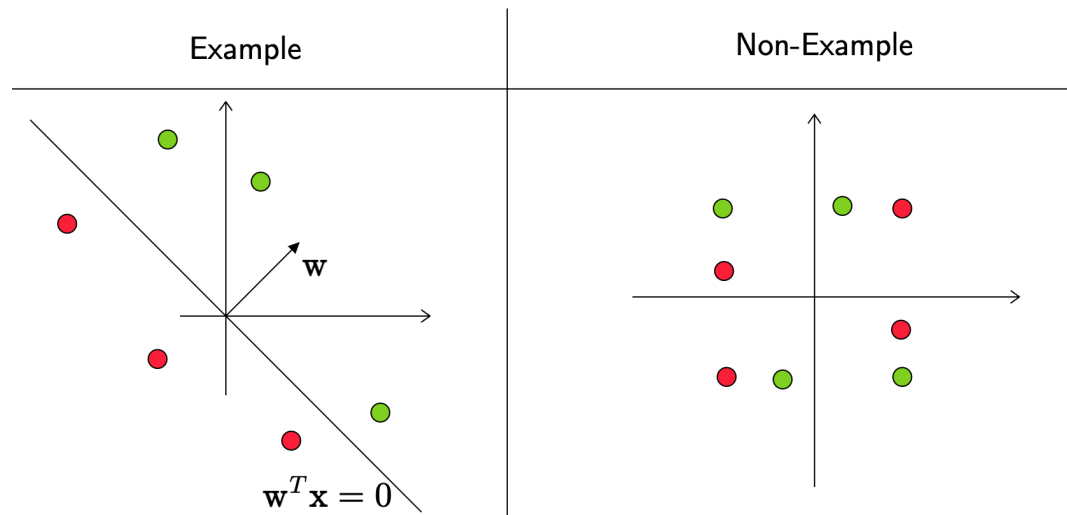
The algorithm is based on two assumption:

1. How to model $P(y = 1 | x)$?

$$P(y = 1|x) = \begin{cases} 1, & \text{if } w^T x > 0 \\ 0, & \text{otherwise} \end{cases}$$

2. Linear Separability Assumption: The Linear Separability Assumption is a fundamental assumption made in various machine learning algorithms, It assumes that the classes to be classified can be accurately separated by a linear decision boundary

$$\exists w \in \mathbb{R}^d \text{ s.t.}$$
$$\text{sign}(w^T x_i) = y_i \forall i \in [n]$$



The objective function is defined as follows

$$\min_{h \in \mathcal{H}} \sum_{i=1}^n \mathbb{1}(h(x_i) \neq y_i)$$

Even if $\mathcal{H}$ accounts only for the Linear Hypotheses, this problem is generally considered NP-Hard. Under the Linear Separability Assumption, assuming the existence of $w \in \mathbb{R}^d$ such that $\text{sign}(w^T x_i) = y_i$ holds for all $i \in \{1, \dots, n\}$, the Perceptron Learning Algorithm solves the convergence problem using an iterative algorithm.

Perceptron Algorithm

Initialise $w^0 = 0 \in \mathbb{R}^d = [0 \ 0 \ . \ . \ 0]$

until convergence

- pick (x_i, y_i) from the dataset

- If $\text{sign}(w^t x_i) = y_i$

do nothing

- else

$w^{t+1} = w^t + x_i \cdot y_i \rightarrow \text{Update Rule}$

end

end

Mistakes by Algorithm

Case I

Predicted = 1 $\rightarrow (w^T x) \geq 0$

Actual = -1

$w^{t+1} = w^t + x_i \cdot y_i$

$$(w^{t+1}) x_i = (w^t + x_i \cdot y_i)^T x_i$$

$$= \underbrace{w^t x_i}_{\geq 0} + \underbrace{y_i}_{-1} \underbrace{\|x_i\|_2^2}_{> 0}$$

Case II

Predicted = -1 $\rightarrow (w^T x) < 0$

Actual = 1

$$w^{t+1} = w^t + x_i \cdot y_i$$

$$\begin{aligned}(w^{t+1})x_i &= (w^t + x_i \cdot y_i)^T x_i \\ &= \underbrace{w^t{}^T x_i}_{< 0} + \underbrace{y_i}_{+1} \underbrace{\|x_i\|_2^2}_{> 0}\end{aligned}$$

This update rule effectively adjusts the decision boundary in the direction of correct classification for the misclassified example. The algorithm is guaranteed to converge to a linearly separable solution if the data is indeed linearly separable. However, if the data is not linearly separable, the perceptron algorithm may not converge to a solution.

Assumptions

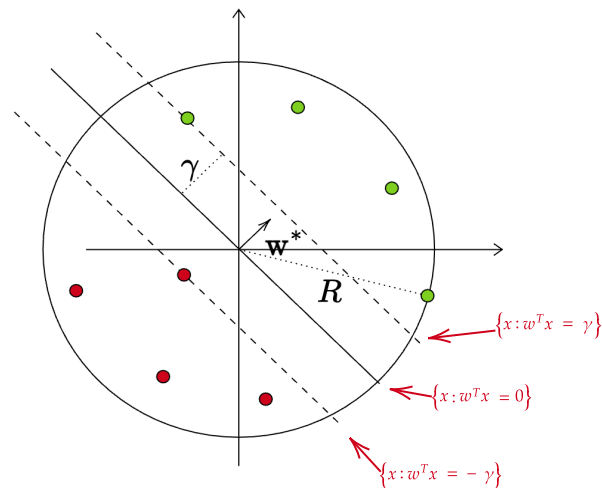
Linear Separability with γ Margin

A dataset $\{(x_1, y_1) \dots (x_n, y_n)\}$ is linearly separable with γ margin

if $\exists w^* \in \mathbb{R}^d$ s.t. $(w^{*T} x_i) y_i \geq \gamma \forall i$ for some $\gamma > 0$

Radius Assumption

$\forall i \in D \quad \|x_i\|_2 \leq R$ for some $R > 0$



Normal Length for w^*

Without Loss of Generality, we assume $\|w^*\| = 1$

$$\left(\frac{w^*}{\|w^*\|} \right)^T x = \frac{\gamma}{\|w^*\|}$$
$$(w^{*'})^T x = \gamma'$$

Analysis of Mistake of Perceptron

Upper bound

Observe that an update happens only when an mistake occurs

Assume that the perceptron has gone through l updates. Let (x, y) be some point that is misclassified by the perceptron at this stage. This is the l^{th} mistake that it is seeing. As the current weight vector of the perceptron is w^l , we have

$$(w^l + x)^T y \leq 0$$

This necessitates one more round of weight update. The update rule for this step is as follows:

$$w^{l+1} = w^l + x \cdot y$$

Let us now look at the norm of the weight vector and see how it changes across iterations:

$$\begin{aligned} \|w^{l+1}\|^2 &= \|w^l + x \cdot y\|^2 \\ &= (w^l + x \cdot y)^T (w^l + x \cdot y) \\ &= \|w^l\|^2 + 2(w^l^T x)y + \|x\|^2 y^2 \\ &\leq \|w^l\|^2 + R^2 \end{aligned}$$

We have used three facts to get the inequality

- $y^2 = 1$
- $\|x\|^2 \leq \mathbb{R}^2$
- $(w^{l^T} x)y \leq 0$

We can now apply this inequality recursively on the earlier rounds

$$\begin{aligned}\|w^{l+1}\|^2 &\leq \|w^l\|^2 + \mathbb{R}^2 \\ &\leq \|w^{l-1}\|^2 + \mathbb{R}^2 + \mathbb{R}^2 \\ &\leq \|w^{l-2}\|^2 + \mathbb{R}^2 + \mathbb{R}^2 + \mathbb{R}^2 \\ &\quad \vdots \\ &\leq \|w^{l-l}\|^2 + l \cdot \mathbb{R}^2 \\ &\leq \|w^0\|^2 + l \cdot \mathbb{R}^2 \\ &\leq 0 + l \cdot \mathbb{R}^2\end{aligned}$$

We have used the fact that $w^{(0)} = 0$ Therefore, the upper bound is:

$$\|w^{l+1}\|^2 \leq l \cdot \mathbb{R}^2$$

Lower bound

Now, we look at the relationship between the optimal weight vector w^* and the weight vector $w^{(t+1)}$. The dot product is a measure of this relationship

$$\begin{aligned}(w^{l+1})^T w^* &= (w^l + x \cdot y)^T w^* \\ &= w^{lT} w^* + (w^{*T} x) y \\ &= w^{lT} w^* + \gamma\end{aligned}$$

We can now apply this inequality recursively on the earlier rounds

$$\begin{aligned}(w^{l+1})^T w^* &\geq w^{lT} w^* + \gamma \\ &\geq (w^{lT} w^* + \gamma) + \gamma \\ &\quad \vdots \\ &\geq (w^{0T} w^* + \gamma) + l \cdot \gamma \\ &\geq 0 + l \cdot \gamma\end{aligned}$$

We have used the fact that $w^{(0)} = 0$. Therefore, the Lower bound is:

$$(w^{l+1})^T w^* \geq l \cdot \gamma$$

Now, we use the Cauchy-Schwartz inequality and the assumption that $\|w^*\| = 1$ to get:

$$\begin{aligned} l \cdot \gamma &\leq (w^{l+1})^T w^* \\ l^2 \cdot \gamma^2 &\leq \left((w^{l+1})^T w^* \right)^2 \leq \|w^{l+1}\|^2 \|w^*\|^2 \\ \|w^{l+1}\|^2 &\geq l^2 \cdot \gamma^2 \end{aligned}$$

Combining the bounds

We now have a lower bound and an upper bound for $\|w^{l+1}\|^2$

$$\begin{aligned} l^2 \cdot \gamma^2 &\leq \|w^{l+1}\|^2 \leq l \cdot \mathbb{R}^2 \\ l^2 \cdot \gamma^2 &\leq l \cdot \mathbb{R}^2 \end{aligned}$$

$$l \leq \frac{\mathbb{R}^2}{\gamma^2}$$

Logistic Regression

Assumption that perceptron makes of linear separability with probabilistic view

$$p(y = 1 | x) = \begin{cases} 1 & \text{if } w^T x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

But, here we are assuming that the probability of some datpoints belongs to positive class is 1 and the probability of some datpoints belongs to negative class is 0, Can we model probabilities differently with some relaxation ?

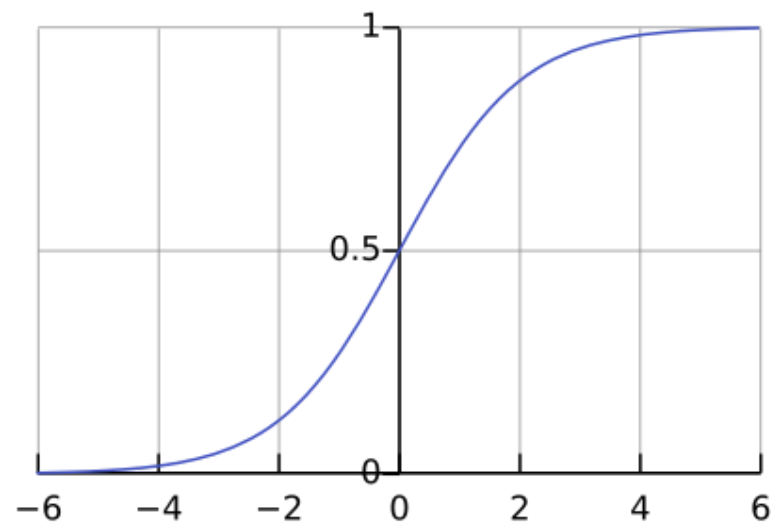
Start with a simple Linear Model

$$z = (w^T x) ; z \in [-\infty, \infty]$$

Larger the value of score ($z = w^T x$) more the probability of being +1

$$\begin{aligned} g(z) &= 0.5, \text{ if } z = 0 \\ g(z) &\rightarrow 1, \text{ as } z \rightarrow \infty \\ g(z) &\rightarrow 0, \text{ as } z \rightarrow -\infty \end{aligned}$$

$$g(z) = \frac{1}{1 + e^{-z}}$$



Model: Logistic Regression

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-w^T x}} = g(w^T x)$$

$$\text{Dataset} = \{(x_1, y_1) \dots (x_n, y_n)\} \quad y_i \in \{0, 1\}$$

How to Find w (Using MLE)

$$\begin{aligned} L(w, \text{data}) &= \prod_{i=1}^n \left(g(w^T x_i) \right)^{y_i} \left(1 - g(w^T x_i) \right)^{(1-y_i)} \\ &= \sum_{i=1}^n y_i \log(g(w^T x_i)) + (1 - y_i) \log(1 - g(w^T x_i)) \\ &= \sum_{i=1}^n \left[(1 - y_i) (-w^T x_i) - \log(1 + e^{-w^T x_i}) \right] \end{aligned}$$

Goal: Maximize $\text{Log}(L(w))$

$$\max_w \sum_{i=1}^n \left[(1 - y_i) (-w^T x_i) - \log(1 + e^{-w^T x_i}) \right]$$

$$\nabla \text{Log}(L(w)) = \sum_{i=1}^n x_i \left(y_i - \frac{1}{1 + e^{-w^T x_i}} \right)$$

Gradient Update Rule

$$\begin{aligned} w^{t+1} &= w^t + \eta_t \nabla \text{Log}(L(w)) \\ &= w^t + \eta_t \left(\sum_{i=1}^n x_i \left(y_i - \frac{1}{1 + e^{-w^T x_i}} \right) \right) \end{aligned}$$

$$\hat{y}_{test} = \text{Sign}(\hat{w}^T x_{test})$$

Kernel Version

Can argue $w^* = \sum_{i=1}^n \alpha_i \cdot x_i$

Regularised Version

$$\min_w \sum_{i=1}^n \left[\log(1 + e^{-w^T x_i}) + (1 - y_i) (w^T x_i) \right] + \frac{\lambda}{2} \|w\|_2^2$$

Reference

Arun Raj Kumar, Lecture Slides for Machine Learning Techniques

Karthik Thiagarajan, Machine Learning Techniques Github Notes.

Indian Institute of Technology Madras , CS2007 Course Website

Stanford University, CS229 Machine Learning Notes

Thank You !